

St George Girls' High School

Trial Higher School Certificate Examination

2002



Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- All questions may be attempted.
- Begin each question on a new page
- All necessary working must be shown.
- Marks may be deducted for careless or poorly presented work.
- Board-approved calculators may be used.
- A list of standard integrals is included at the end of this paper.
- The mark allocated for each question is listed at the side of the question.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 – (12 marks) – Start a new page

Marks

- a) Solve for x

$$\frac{5}{x+3} \leq 1$$

2

- b) Find the coordinates of the point that divides the interval AB with $A(-4,8)$ and $B(6,3)$ in the ratio 3:2.

2

- c) Find the exact value of $\int_0^{\sqrt{2}} \frac{dx}{\sqrt{4-x^2}}$

2

- d) Find the remainder when the polynomial $P(x) = 2x^3 - 3x$ is divided by $x+2$.

2

- e) Find $\frac{d}{dx}\left(\frac{\ln x}{x}\right)$ and hence evaluate $\int_1^2 \frac{\ln x}{x^2} dx$

4

Question 2 – (12 marks) – Start a new page

Marks

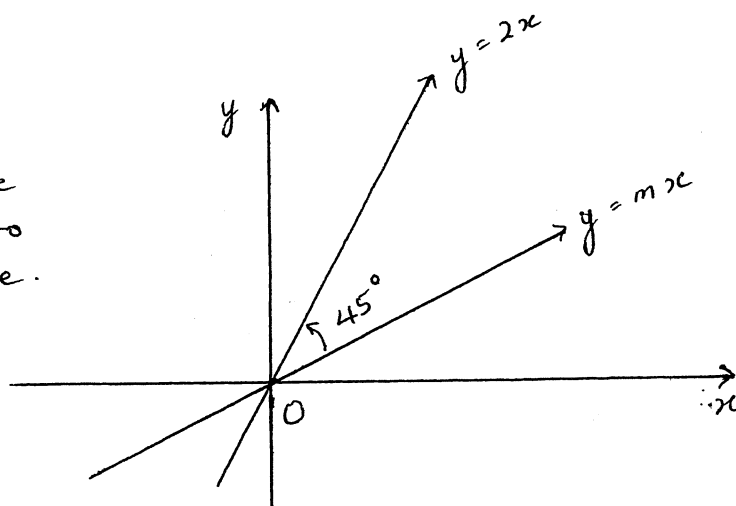
- a) Show that the circle $x^2 + y^2 + 6x - 10y + 25 = 0$ touches the y -axis and give the coordinates of the point of contact. 2
- b) A particle is moving in simple harmonic motion. Its displacement x at any time t is given by $x = 3 \cos(2t + 5)$. 5
- (i) Find the period of the motion.
- (ii) Find the maximum acceleration of the particle.
- (iii) Find the speed of the particle when $x = 2$.
- c) (i) Write down the expansion for $\tan(A - B)$ and use this to deduce that the acute angle θ between two lines is given by:

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

5

(ii)

Figure
not to
scale.



The angle between the lines $y = 2x$ and $y = mx$ is 45° as shown in the diagram. Find the exact value of m .

Question 3 – (12 marks) – Start a new page

Marks

a)

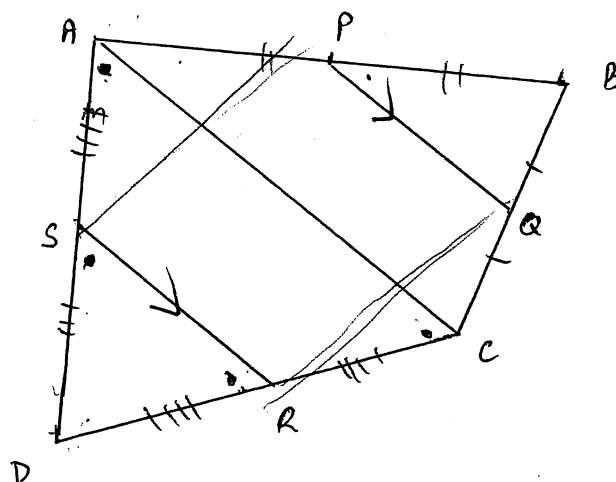


Figure not
to scale.

The sides of a quadrilateral $ABCD$ have midpoints P , Q , R and S as shown.

(i) Show $\triangle DSR$ is similar to $\triangle DAC$.

(ii) Show $RS \parallel QP$.

(iii) Show $PQRS$ is a parallelogram.

b) Consider the function $f(x) = 2 \tan^{-1} x$.

(i) Find the exact value of $f(\sqrt{3})$.

(ii) Find the equation of the tangent to the curve at the point where $x = \sqrt{3}$.

c) A pool holds a volume of water given by $V = 3x + 2x^2$, where x is the depth of the water. If the pool is filled with water at the rate of 0.9 m^3 per hour at what rate will the level of water be increasing when the depth is 1.2 m .

Question 4 – (12 marks) – Start a new page

Marks

- a) (i) Express $\sqrt{3} \cos x - \sin x$ in the form $R \cos(x + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq 2\pi$.

4

- (ii) Hence find the general solution of $\sqrt{3} \cos x - \sin x = 1$.

- b) The function $f(x) = x^2 - \ln(x+1)$ has one root between 0.5 and 1.

4

- (i) Show that the root lies between 0.7 and 0.8.

- (ii) Hence using halving-the-interval method find the value of the root correct to one decimal place.

- c) Paula walks along a straight road. At one point she notices a tower on a bearing of 055° with an angle of elevation of 23° . After walking 240m the tower is on a bearing of 345° with an angle of elevation of 27° .

4

- (i) Draw a diagram to represent the above information.

- (ii) Show that the height (h) of the tower is given by

$$h^2 = \frac{240^2}{\cot^2 23^\circ + \cot^2 27^\circ - 2 \cot 23^\circ \cot 27^\circ \cos 70^\circ}$$

- (iii) Calculate the height of the tower to the nearest metre.

Question 5 – (12 marks) – Start a new page

Marks

- a) Consider the equation $x^3 + 2x^2 - 19x - 20 = 0$. One of the roots of this equation is equal to the sum of the other two roots.

Find the values of the three roots.

4

- b) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on $x^2 = 4ay$.
The normals at P and Q intersect at M

4

- (i) Show the equation of the normal at P is $x + py = 2ap + ap^3$.

- (ii) Show that the coordinates of M are $(-apq^2 - ap^2q, 2a + a(p^2 + q^2 + pq))$.

- c) Yasmin invests $\$P$ at 6% p.a. compounded annually. She plans to withdraw $\$4000$ at the end of each year for the next 4 years to cover her son's university fees.

4

- (i) Write down the amount $\$A_1$ remaining in the account following the first withdrawal.

- (ii) Find an expression for the amount $\$A_2$ remaining in the account after the second withdrawal.

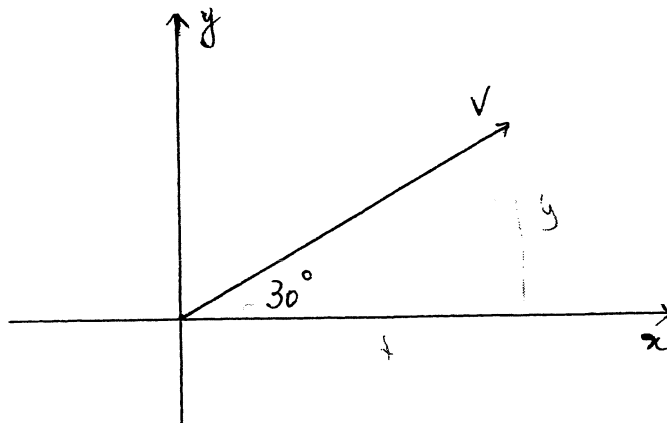
- (iii) Calculate the amount Yasmin needs to invest if the account balance is to be $\$0$ after 4 years.

Question 6 – (12 marks) – Start a new page

Marks

a)

8



A particle is projected at an angle of 30° with a velocity of v metres per second. The equations of motion of the particle are:

$$\ddot{x} = 0 \quad \text{and} \quad \ddot{y} = -g$$

- (i) Using calculus, derive the expression for the position of the particle at time t . Hence, show the path of the particle is given by:

$$y = \frac{x}{\sqrt{3}} - \frac{2g}{3v^2} x^2$$

A golfer hitting off from the 4th tee, strikes a ball with initial speed $v \text{ ms}^{-1}$ and an angle of projection of 30° . The ball just clears a 3m bush which is 120m from the golfer.

- (ii) Show that the initial speed of the ball is approximately 37.7 ms^{-1} (take $g = 9.8 \text{ ms}^{-2}$).

- (iii) What is the horizontal distance from the bush to the point where the ball lands?

- b) Prove by induction that $3^{3n} + 2^{n+2}$ is divisible by 5 for all integers $n \geq 1$.

4

Question 7 – (12 marks) – Start a new page

Marks

- a) (i) Sketch the function $f(x) = \cos^{-1} x$ 2
- (ii) Find the exact value of $f\left(\frac{1}{2}\right)$. 1
- (iii) Find the exact area bounded by the curve $y = f(x)$, the x -axis and the lines $x = 0$ and $x = \frac{1}{2}$. 3
- (iv) Find the volume formed if the curve $f(x) = \cos^{-1} x$ is rotated about the x -axis between $x = 0$ and $x = \frac{1}{2}$, using Simpson's Rule with 3 function values. Give your answer correct to 2 decimal places. 2
- b) The acceleration a metres per second of a particle P moving in a straight line is given by $a = 1 - 9x^2$ where x metres is the displacement of the particle to the right of the origin. Initially the particle is at the origin moving with a velocity of 4 ms^{-1} . 4
- (i) Show that the velocity $v \text{ ms}^{-1}$ of the particle is given by $v^2 = 16 + 2x - 6x^3$.
- (ii) Will the particle ever return to the origin? Justify your answer.

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Q1

a) $\frac{5}{x+3} \leq 1$ $x = -3$

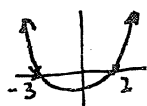
$$5(x+3) < (x+3)^2$$

$$(x+3)^2 - 5(x+3) > 0$$

$$(x+3)(x+3-5) > 0$$

$$(x+3)(x-2) > 0$$

$$x < -3 \quad x > 2 \quad (2)$$



$$\begin{aligned} &= - \left[\frac{\ln x}{x} + x^{-1} \right]_1^2 \\ &= - \left(\frac{\ln 2}{2} + \frac{1}{2} \right) + (0+1) \\ &= - \frac{\ln 2}{2} + \frac{1}{2} \\ &= \frac{1 - \ln 2}{2} \quad (4) \end{aligned}$$

b) $A(-4, 8)$ $B(6, 3)$ $m:n = 3:2$

$$x = \frac{mx_2 + nx_1}{m+n} \quad y = \frac{my_2 + ny_1}{m+n}$$

$$= \frac{3 \times 6 + 2 \times -4}{5} \quad = \frac{3 \times 3 + 2 \times 8}{5}$$

$$= 2$$

$$= 5$$

$$(2, 5)$$

(2)

c) $\int_0^{\sqrt{2}} \frac{dx}{\sqrt{4-x^2}} = \left[\sin^{-1} \frac{x}{2} \right]_0^{\sqrt{2}}$

$$\begin{aligned} &= \sin^{-1} \frac{\sqrt{2}}{2} - \sin^{-1} 0 \\ &= \frac{\pi}{4} \end{aligned}$$

(2)

d) $P(x) = 2x^3 - 3x$

$$P(-2) = 2(-2)^3 - 3(-2)$$

$$= -16 + 6$$

$$= -10$$

e) $\frac{d}{dx} \left(\frac{\ln x}{x} \right) = \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2}$

$$= \frac{1 - \ln x}{x^2}$$

$$\int_1^2 \frac{\ln x}{x^2} dx$$

$$= \int_1^2 \frac{-\ln x + 1 - 1}{x^2} dx$$

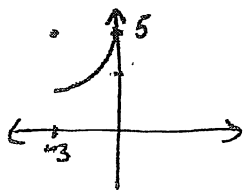
$$= - \int_1^2 \frac{1 - \ln x}{x^2} - x^{-2} dx$$

Q2 a) $x^2 + y^2 + 6x - 10y + 25 = 0$

$x^2 + 6x + 9 + y^2 - 10y + 25 = -25 + 34$ c)

$(x+3)^2 + (y-5)^2 = 9$

circle centre $(-3, 5)$ $r = 3$ (2)



$(0, 5)$ is the
pt of contact
3 units from $(-3, 5)$

ii) $\tan 45^\circ = \left| \frac{2-m}{1+2m} \right|$ for acute angle

$1 = \left| \frac{2-m}{1+2m} \right|$

$1+2m = 2-m$

$1+2m = -2+m$

$3m = 1$

$m = -3$

$m = \frac{1}{3}$

b) i) $x = 3 \cos(2t+5)$ $T = \frac{2\pi}{\omega}$

$T = \frac{2\pi}{2}$
 $= \pi$

$n = 2$

ii) $\dot{x} = -6 \sin(2t+5)$

$\ddot{x} = -12 \cos(2t+5)$

max when $\cos(2t+5) = -1$

$\ddot{x} = 12$

iii) $x = 2$

$\ddot{x} = -\omega^2 x$

$\ddot{x} = \frac{d}{dt} \left(\frac{1}{2} v^2 \right)$

$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -4x$

$\frac{1}{2} v^2 = -2x^2 + \frac{1}{2} C$

$v^2 = -4x^2 + C$

$C = 36$

$v = 0, x = 3$

$v^2 = -4x^2 + 36$

$v^2 = -16 + 36$

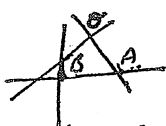
$x = 2$

$= 20$

$v = \sqrt{20}$

speed = $2\sqrt{5}$ m/s.

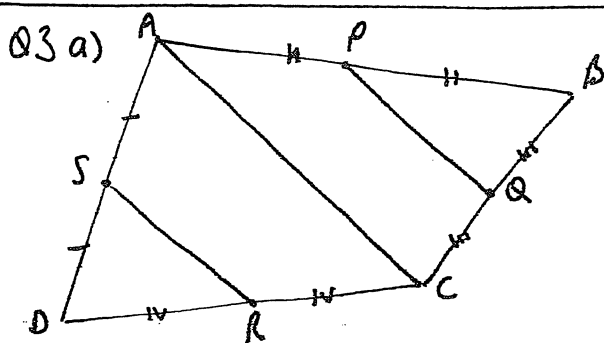
c) i) $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$



$A = \theta + \phi$

$\theta = A - \phi$

$= \frac{m_1 - m_2}{1 + m_1 m_2}$



i) In Δ 's ΔSR ΔDAC

$$DS : DA = 1 : 2$$

$$DR : DC = 1 : 2$$

$\hat{D}$ is common

$\therefore \Delta SR \sim \Delta DAC$ (2 pairs of sides)

in the same ratio and the included angle equal.)

ii) $\Delta SR \sim \Delta DAC$

$$\hat{DSR} = \hat{DAC} \text{ (corr } \angle \text{'s of sim } \Delta \text{'s)}$$

$$\therefore SR \parallel AC \text{ (corr } \angle \text{'s equal)}$$

$$\hat{BPQ} = \hat{BAC} \text{ (corr } \angle \text{'s of sim } \Delta \text{'s)}$$

$$\therefore PQ \parallel AC \text{ (corr } \angle \text{'s equal)}$$

$$\therefore RS \parallel QP \text{ (parallel to same line)}$$

$$(iii) PQ : AC = 1 : 2 \text{ (corr sides in sim } \Delta \text{'s)}$$

$$SR : AC = 1 : 2 \text{ (corr sides in sim } \Delta \text{'s)}$$

$$\therefore PQ : SR = 1 : 1$$

$$\therefore PQ = SR$$

$\therefore PQRS$ is a parallelogram
(1 pair of sides equal and parallel)

$$b) i) f(x) = 2 \tan^{-1} x$$

$$f(\sqrt{3}) = 2 \tan^{-1} \sqrt{3}$$

$$= 2 \cdot \frac{\pi}{3}$$

$$= \frac{2\pi}{3}$$

$$ii) f'(x) = \frac{2}{1+x^2} \quad \text{at } x = \sqrt{3}$$

$$= \frac{2}{4}$$

$$= \frac{1}{2}$$

$$3bii) y - \frac{2\pi}{3} = \frac{1}{2}(x - \sqrt{3})$$

$$2y - \frac{4\pi}{3} = x - \sqrt{3}$$

$$x - 2y - \sqrt{3} + \frac{4\pi}{3} = 0$$

$$c) V = 3x + 2x^2 \quad \frac{dV}{dt} = 0.9$$

$$\frac{dV}{dt} = \frac{dV}{dx} \cdot \frac{dx}{dt} \quad x = 1.2$$

$$0.9 = (3 + 4x) \frac{dx}{dt}$$

$$0.9 = 7.8 \frac{dx}{dt}$$

$$\frac{dx}{dt} = \frac{3}{26} \text{ m/s.}$$

$$= 0.115 \text{ m/s}$$

Q4 a)

$$\begin{aligned}\sqrt{3} \cos x - \sin x &= R \cos(x+d) \\ &= R \cos x \cos d - R \sin x \sin d\end{aligned}$$

$$\begin{aligned}R \cos d &= \sqrt{3} & R \sin d &= 1 \\ \frac{R \sin d}{R \cos d} &= \frac{1}{\sqrt{3}} & R &= \sqrt{3+1} \\ & & &= 2.\end{aligned}$$

$$\begin{aligned}\tan d &= \frac{1}{\sqrt{3}} \\ d &= \frac{\pi}{6}, \quad \frac{11\pi}{6}\end{aligned}$$

$$\begin{aligned}\therefore 2 \cos\left(x + \frac{\pi}{6}\right) &= 1 \\ \cos\left(x + \frac{\pi}{6}\right) &= \frac{1}{2} \\ x + \frac{\pi}{6} &= \frac{\pi}{3} \\ x + \frac{\pi}{6} &= 2n\pi \pm \frac{\pi}{3} \\ x &= 2n\pi + \frac{\pi}{6}, 2n\pi - \frac{\pi}{2}\end{aligned}$$

b) $f(x) = x^2 - \ln(x+1)$

i) $f(0.7) = 0.7^2 - \ln(1.7)$
 $= -0.040628251$

$f(0.8) = 0.8^2 - \ln(1.8)$
 $= 0.052213335$

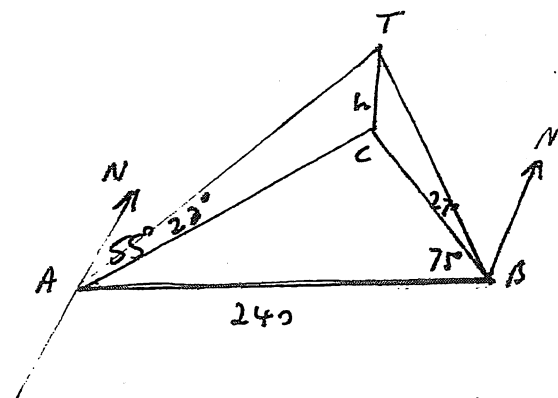
$\therefore$ root lies between 0.7 and 0.8
as there is a change in sign.

ii)

x	0.7	0.8	0.75	0.725	0.7375	0.74
y	-0.04	0.05	0.002	-0.01	-0.008	

$\therefore 0.7$ (to 1 dp)

c)



$$\begin{aligned}\text{In } \triangle ATC \quad \tan 23^\circ &= \frac{h}{AC} \\ \triangle BTC \quad \tan 27^\circ &= \frac{h}{BC}\end{aligned}$$

In $\triangle ACB$
 $240^2 = AC^2 + BC^2 - 2 \times AC \times BC \cos(180 - 35 - 75)$

$$\begin{aligned}4c) \quad 240^2 &= \left(\frac{h}{\tan 23^\circ}\right)^2 + \left(\frac{h}{\tan 27^\circ}\right)^2 \\ &\quad - 2 \times \frac{h}{\tan 23^\circ} \times \frac{h}{\tan 27^\circ} \cos 70^\circ\end{aligned}$$

$$240^2 = \frac{h^2}{\tan^2 23^\circ} + \frac{h^2}{\tan^2 27^\circ} - \frac{2h^2 \cos 70^\circ}{\tan 23^\circ \tan 27^\circ}$$

$$\begin{aligned}240^2 &= h^2 \cot^2 23^\circ + h^2 \cot^2 27^\circ - 2h^2 \cos 70^\circ \cot 23^\circ \cot 27^\circ \\ &= h^2 (\cot^2 23^\circ + \cot^2 27^\circ - 2 \cos 70^\circ \cot 23^\circ \cot 27^\circ)\end{aligned}$$

$$\therefore h^2 = \frac{240}{\cot^2 23^\circ + \cot^2 27^\circ - 2 \cos 70^\circ \cot 23^\circ \cot 27^\circ}$$

$$\begin{aligned}\text{ii) } h &= 96.0835321 \\ &= 96 \text{ to nearest m.} \\ \text{height is } 96 \text{ m.}\end{aligned}$$

Q5

$$a) x^3 + 2x^2 - 19x - 20 = 0$$

$$\alpha, \beta, \alpha + \beta$$

$$\alpha + \beta + \alpha + \beta = -\frac{b}{a}$$

$$2\alpha + 2\beta = -2$$

$$\alpha + \beta = -1$$

$$\alpha(\alpha + \beta) + \alpha\beta + \beta(\alpha + \beta) = \frac{c}{a}$$

$$\alpha^2 + \alpha\beta + \alpha\beta + \beta^2 = -19$$

$$\alpha^2 + 3\alpha\beta + \beta^2 = -19$$

$$(\alpha + \beta)^2 + \alpha\beta = -19$$

$$\alpha\beta = -20$$

$$\alpha = -\frac{20}{\beta}$$

$$-\frac{20}{\beta} + \beta = -1$$

$$-20 + \beta^2 = -\beta$$

$$\beta^2 + \beta - 20 = 0$$

$$(\beta + 5)(\beta - 4) = 0$$

$$\beta = -5, 4$$

$$\alpha = 4, \beta = -5 \quad \gamma = -1$$

$$i) P(2ap, ap^2) \quad Q(2aq, aq^2)$$

$$x^2 = 4ay$$

$$y = \frac{x^2}{4a}$$

$$y' = \frac{2x}{4a}$$

$$= \frac{x}{2a} \quad \text{at } 2ap$$

$$y' = p$$

$$\therefore \text{grad of normal is } -\frac{1}{p}$$

$$\therefore y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$x + py = 2ap + ap^3$$

5b) ii) Similarly

eqⁿ at Q is

$$x + qy = 2aq + aq^3$$

$$2ap + ap^3 - py + qy = 2aq + aq^3$$

$$(q - p)y = 2aq + aq^3 - 2ap - ap^3$$

$$= 2a(q - p) + a(q^3 - p^3)$$

$$y = 2a + a(q^2 + qp + p^2)$$

$$x + p(2a + a(q^2 + qp + p^2)) = 2ap + ap^3$$

$$x = 2ap + ap^3 - 2ap - apq^2 - aqp^2 - ap^3$$

$$= -apq^2 - aqp^2$$

$$\therefore M(-apq^2 - aqp^2, 2a + a(p^2 + q^2 + pq))$$

c) \$P 6\% p.a annually $M = 4000$

$$n = 4$$

$$i) A_1 = P \times 1.06 - 4000$$

$$ii) A_2 = (P \times 1.06 - 4000) \times 1.06 - 4000$$

$$= P \times 1.06^2 - 4000 \times 1.06 - 4000$$

$$= P \times 1.06^2 - 4000(1.06 + 1)$$

$$iii) A_3 = P \times 1.06^3 - 4000(1.06^2 + 1.06) - 4000$$

$$= P \times 1.06^3 - 4000(1.06^2 + 1.06 + 1)$$

$$A_4 = P \times 1.06^4 - 4000(1.06^3 + 1.06^2 + 1.06 + 1)$$

$$A_4 = 0$$

$$\therefore P \times 1.06^4 - 4000(1 + 1.06 + 1.06^2 + 1.06^3) = 0$$

$$G \& a = 1$$

$$r = 1.06$$

$$n = 4$$

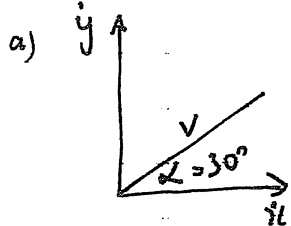
$$P \times 1.06^4 = \frac{4000 \times 1(1.06^4 - 1)}{1.06 - 1}$$

$$P = \frac{4000 \times (1.06^4 - 1)}{1.06^4 \times 0.06}$$

$$= 66666.67 \$ \quad 13860.42$$

$\therefore$ Principal \$66666.67

Q6



$$\begin{aligned} \dot{y} &= v \sin 30 \\ &= \frac{v}{2} \\ \dot{x} &= v \cos 30 \\ &= \frac{\sqrt{3}v}{2} \end{aligned}$$

$$\begin{aligned} \text{i) } \ddot{x} &= 0 \\ \dot{x} &= C_1 & \dot{x} &= \frac{\sqrt{3}v}{2} \\ \dot{x} &= \frac{\sqrt{3}v}{2} \end{aligned}$$

$$x = \frac{\sqrt{3}vt}{2} + C_2 \quad \begin{matrix} t=0 \\ x=0 \\ C_2=0 \end{matrix}$$

$$\therefore x = \frac{\sqrt{3}vt}{2}$$

$$t = \frac{2x}{\sqrt{3}v}$$

$$\text{ii) } x = 120 \text{ m} \quad y = 3 \quad g = 9.8$$

$$3 = \frac{120}{\sqrt{3}} - \frac{19.6}{3v^2} (120)^2$$

$$\frac{19.6}{3v^2} (120)^2 = \frac{120}{\sqrt{3}} - 3$$

$$v^2 = \frac{19.6 \times 120^2}{3 \times \left(\frac{120}{\sqrt{3}} - 3 \right)}$$

$$= 1419.389$$

$$v = 37.7 \text{ m/s}$$

b) Step 1 Prove true for $n=1$

$$\text{LHS} = 3^3 + 2^3$$

$$= 27 + 8$$

$$= 35$$

$$= 5 \times 7$$

$\therefore$ True for $n=1$

$$y = -gt^2$$

$$\dot{y} = -gt + C_3$$

$$\dot{y} = -gt + \frac{v}{2} \quad t=0 \quad \therefore C_3 = \frac{v}{2}$$

$$y = -\frac{gt^2}{2} + \frac{vt}{2} + C_4 \quad y=0 \quad t=0 \quad \therefore C_4 = 0$$

$$\therefore y = -\frac{gt^2}{2} + \frac{vt}{2}$$

$$\begin{aligned} \therefore y &= -g \left(\frac{2x}{\sqrt{3}v} \right)^2 + \frac{v}{2} \left(\frac{2x}{\sqrt{3}v} \right) \\ &= -g \frac{4x^2}{6v^2} + \frac{x}{\sqrt{3}} \\ &= \frac{x}{\sqrt{3}} - \frac{2g}{3v^2} x^2 \end{aligned}$$

$$\text{iii) } y = 0$$

$$0 = \frac{x}{\sqrt{3}} - \frac{2 \times 9.8}{3 \times 1419.389} x^2$$

$$= 3 \times 1419.389 x - 2 \times 9.8 x \sqrt{3}$$

$$= x (3 \times 1419.389 - 2 \times 9.8 \times \sqrt{3})$$

$$\therefore x=0 \quad x = \frac{3 \times 1419.389}{2 \times 9.8 \times \sqrt{3}}$$

$$x = 125.43$$

$\therefore$ 5.4 metres from bus

Step 2

Assume true for $n=k$

i.e. $3^{3k} + 2^{k+2} = 50$ for all $n \geq 1$

Prove true for $n=k+1$

i.e. $3^{3(k+1)} + 2^{(k+1)+2} = 5K$

$$\begin{aligned} \text{LHS} &= 3^{3(k+1)} + 2^{(k+1)+2} \\ &= 3^{3k+3} + 2^{k+3} \\ &= 3^3 (3^{3k}) + 2^{k+3} \\ &= 3^3 (3^{3k} + 2^{k+2} - 2^{k+2}) + 2^{k+3} \\ &= 27 (3^{3k} + 2^{k+2}) - 27 \cdot 2^{k+2} + 2 \cdot 2^{k+2} \\ &= 27 \times 50 - 25 \times 2^{k+2} \\ &= 5 (270 - 5 \cdot 2^{k+2}) \end{aligned}$$

$\therefore$ divisible by 5.

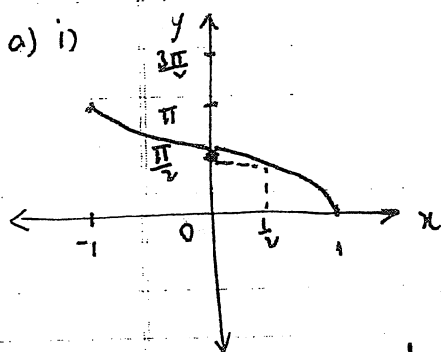
Therefore if the assertion is true for $n=k$ then it is also true for $n=k+1$

Step 3.

Since assertion is true for $n=1$ then it is true for $n=2$ and by Induction it is true for all $n \geq 1$

Q7.

a) i)



$$ii) f\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

$$iii) \text{Area} = \int_0^{1/2} \cos^{-1} x \, dx$$

$$y = \cos^{-1} x \\ x = \cos y$$

$$= \int_{\pi/3}^{\pi/2} x \, dy + \text{rect}$$

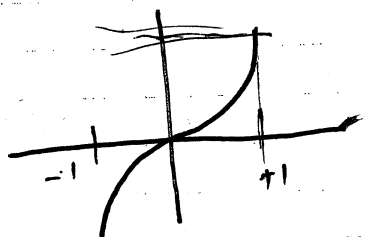
$$= \int_{\pi/3}^{\pi/2} \cos y \, dy + \frac{1}{2} \times \frac{\pi}{3}$$

$$= \left[\sin y \right]_{\pi/3}^{\pi/2} + \frac{\pi}{6}$$

$$= 1 - \sin \frac{\pi}{3} + \frac{\pi}{6}$$

$$= 1 - \frac{\sqrt{3}}{2} + \frac{\pi}{6}$$

$$\int_0^1 \sin^{-1} x \, dx$$



iv)

x	0	$\frac{1}{4}$	$\frac{1}{2}$
$f(x)$	$\frac{\pi}{2}$	1.318	$\frac{\pi}{3}$

$$(f(x))^2 \quad 2.467 \quad 1.737 \quad 1.047$$

$$1 \times \quad 4 \times \quad 1 \times$$

$$x = \sin y$$

$$\begin{aligned} \text{Volume} &= \frac{1}{6} \left[1 \times 2.467 + 4 \times 1.737 + 1 \times 1.047 \right] \\ &= 1.88558333 \times \pi \\ &= 1.90 \times \pi \end{aligned}$$

$$v) i) a = 1 - 9x^2 \quad t=0, x=0, v=4$$

$$ii) \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 1 - 9x^2$$

$$\frac{1}{2} v^2 = x - 3x^3 + C \quad v=4, x=0$$

$$8 = C$$

$$C = 8$$

$$\therefore \frac{1}{2} v^2 = x - 3x^3 + 8$$

$$v^2 = 2x - 6x^3 + 16$$

$$v^2 = 16 + 2x - 6x^3$$

$$ii) x=0, v^2=16 \quad v=\pm 4 \quad \text{we know it is } 4$$

$$x=1, v^2=12 \quad v=\pm 2\sqrt{3}$$

$$x=2, v^2=-28$$

$$\therefore \text{Between } x=1 \text{ and } x=2, v=0$$

$\therefore$ velocity changes to negative

$\therefore$ particle returns to zero

$$\text{as } a=0 \text{ at } x=\pm \frac{1}{3}$$

there is a local minimum at $x=-\frac{1}{3}$

$\therefore x$ doesn't return to zero

